FPT SCHOOL OF BUSINESS
& TECHNOLOGY

SAMPLING OF CONTINUOUS-TIME SIGNALS

OBJECTIVES

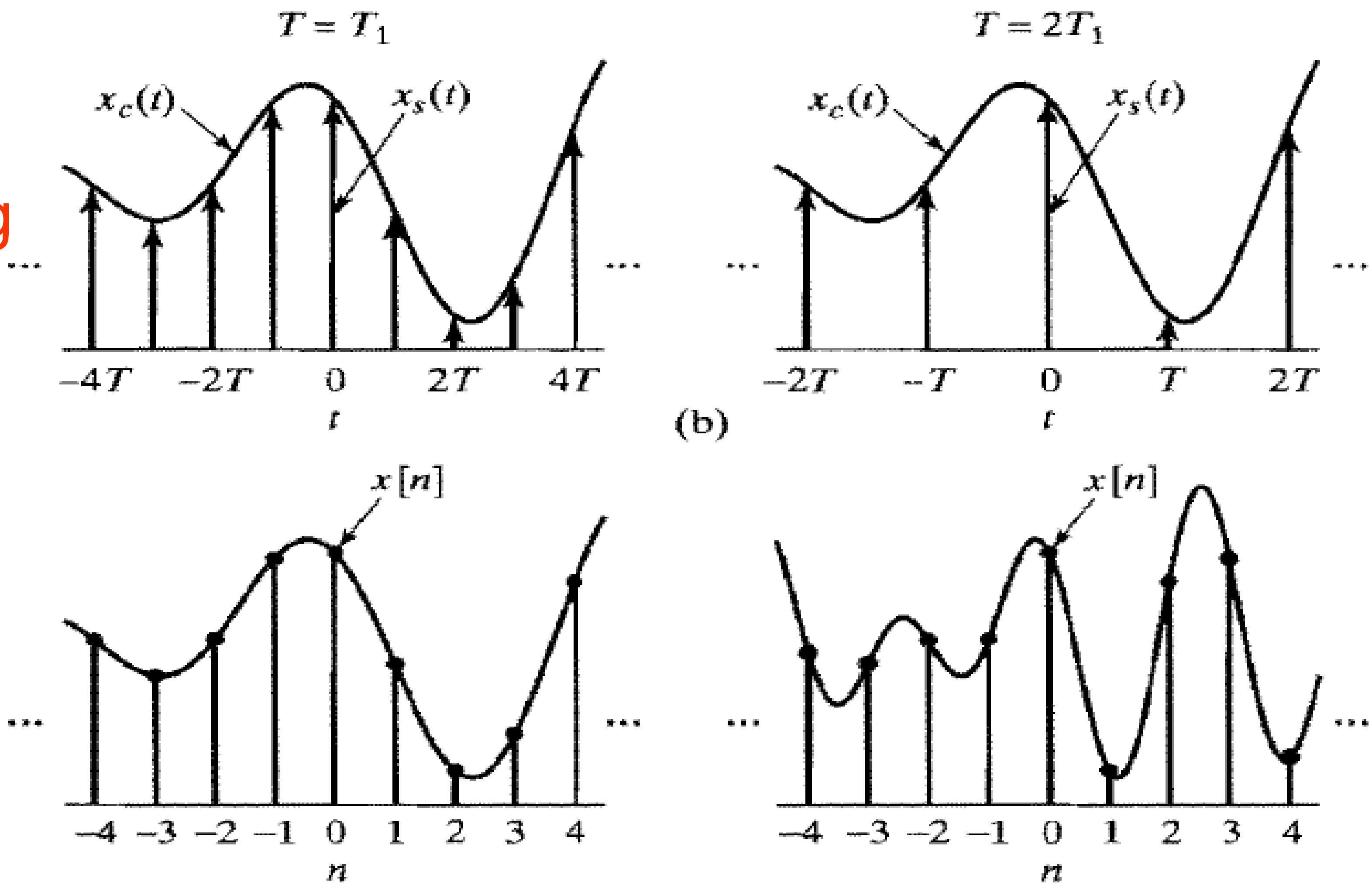
- Understand the sampling
- Know the ways to reconstruct signals
- Understand to process signals

SAMPLING OF CONTINUOUS-TIME SIGNALS

1. Periodic Sampling
2. Frequency-Domain Representation of Sampling
3. Reconstruction of a Bandlimited Signal from its Samples
4. Discrete-Time Processing of Continuous-Time signals
5. Continuous-time processing of discrete-time signals

PERIODIC SAMPLING

T: sampling period



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FREQUENCY-DOMAIN REPRESENTATION OF SAMPLING

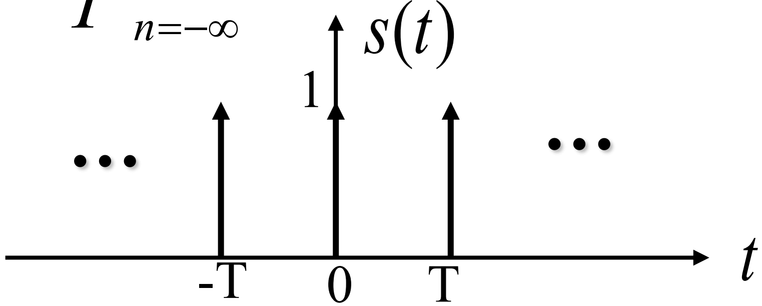
https://aita-lab.github.io

$$S(j\Omega)=\frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\delta(\Omega-k\Omega_s)$$

T： sample period; $f_s=1/T$:sample rate;$\Omega_s=2\pi/T$:sample rate

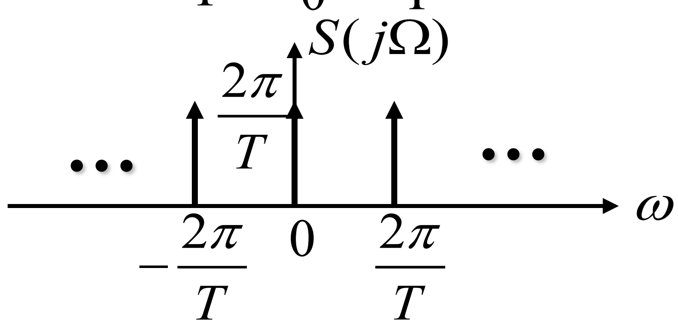
$$s(t)=\sum_{n=-\infty}^{\infty}\delta(t-nT) \quad =\sum_{n=-\infty}^{\infty}a_k e^{jk\Omega_s t} \quad =\frac{1}{T}\sum_{n=-\infty}^{\infty}e^{jk\Omega_s t}$$

$$a_k=\frac{1}{T}\int_{-T/2}^{+T/2}\delta(t)e^{-jk\Omega_s t}dt=\frac{1}{T}$$



$$e^{jk\Omega_s t}\xleftrightarrow{F}2\pi\delta(\Omega-k\Omega_s)$$

$$S(j\Omega)=\frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\delta(\Omega-k\Omega_s)$$

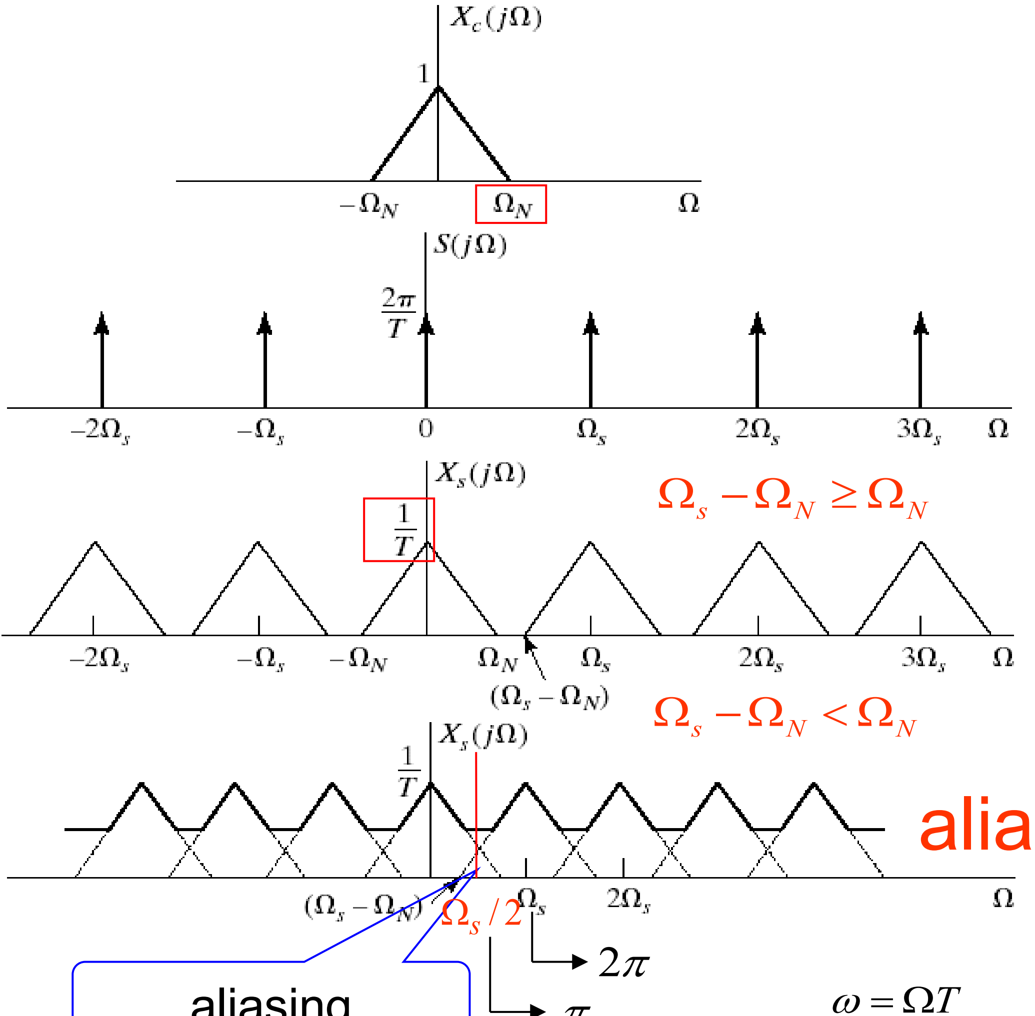


FREQUENCY-DOMAIN REPRESENTATION OF SAMPLING

•Nyquist Sampling Theorem

- Let $X_c(t)$ be a bandlimited signal with
  . $X_c(j\Omega) = 0 \quad for\ |\Omega| \geq \Omega_N$
- Then $X_c(t)$ is uniquely determined by its samples
  $$x[n] = x_c(nT), \quad n = 0, \pm 1, \pm 2, \ldots$$
  ,if $\Omega_s = \dfrac{2\pi}{T} \geq 2\Omega_N$
- The frequency Ω_N is commonly referred as the *Nyquist frequency*.
- The frequency $2\Omega_N$ is called the *Nyquist rate*.

FREQUENCY-DOMAIN REPRESENTATION OF SAMPLING

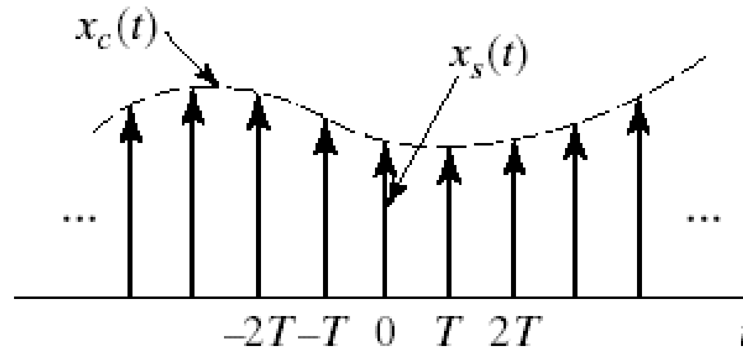


$\Omega_s - \Omega_N \geq \Omega_N$

$\Omega_s - \Omega_N < \Omega_N$

aliasing frequency

$\omega = \Omega T$



frequency spectrum of ideal sample signal

$$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

No aliasing

aliasing

$$X(e^{j\omega}) = X_s(j\Omega)\big|_{\Omega=\omega/T}$$

$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\omega - k2\pi)/T)$$

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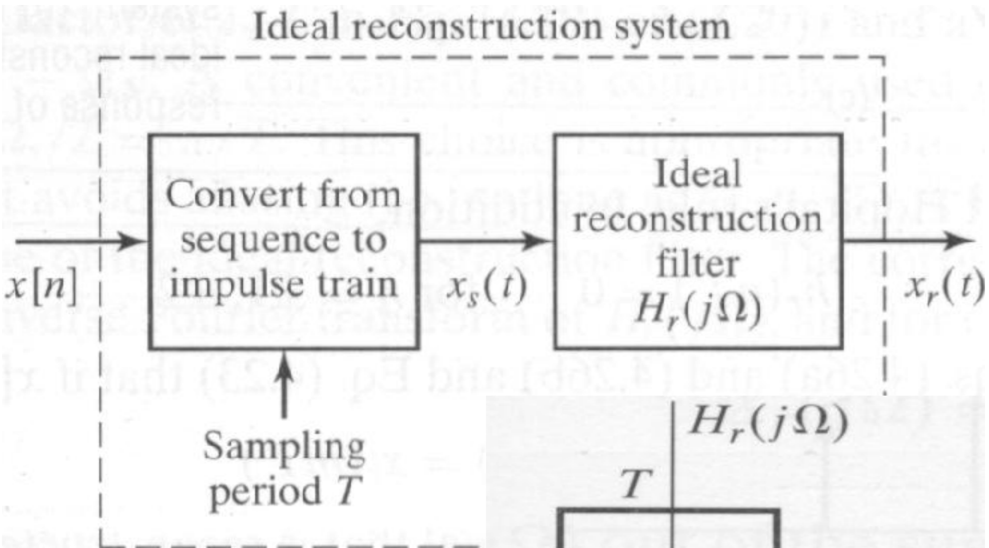
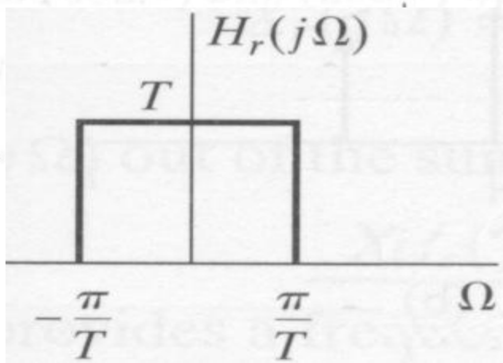
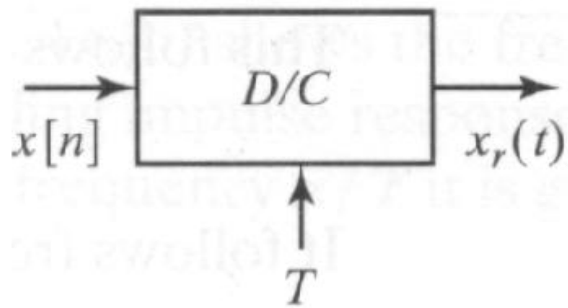
RECONSTRUCTION OF A BANDLIMITED SIGNAL





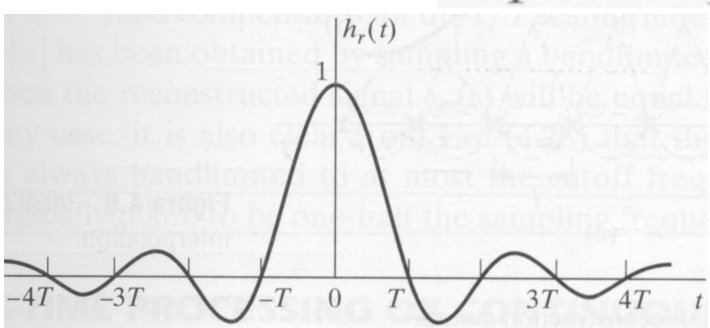
Gain: T



$$h_r(t) = \frac{\sin(\pi t/T)}{\pi t/T}$$

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] h_r(t - nT)$$

$$= \sum_{n=-\infty}^{\infty} x[n] \frac{\sin\left[\pi(t - nT)/T\right]}{\pi(t - nT)/T}$$



$$X_r(j\Omega) = H_r(j\Omega) X\left(e^{j\Omega T}\right)$$

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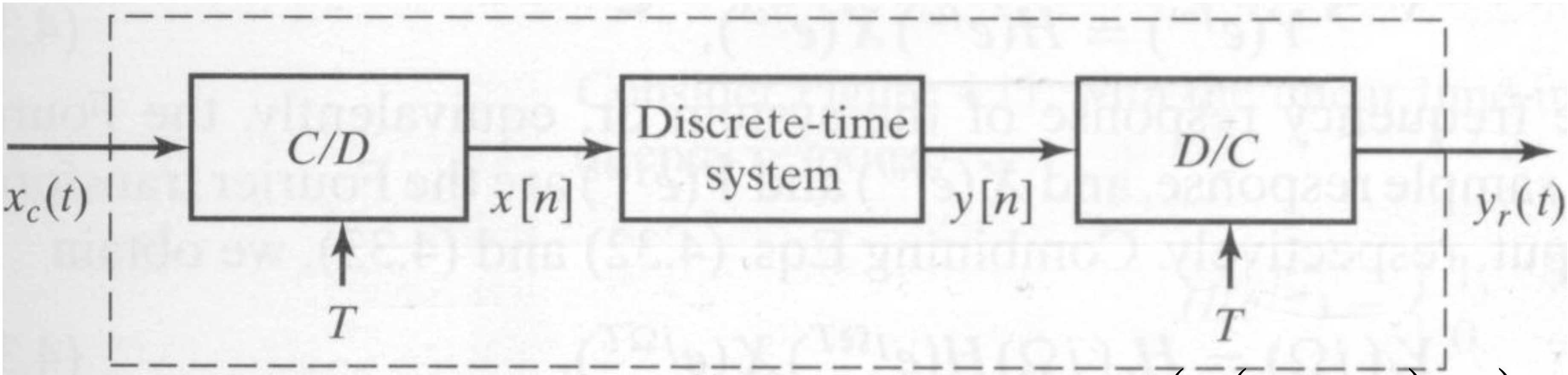

$$x[n] = x_c(nT)$$

$$y_r(t) = \sum_{n=-\infty}^{\infty} y[n] \frac{\sin(\pi(t-nT)/T)}{\pi(t-nT)/T}$$

$$X(e^{jw}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{w}{T} - \frac{2\pi k}{T}\right)\right)$$

$$Y_r(j\Omega) = H_r(j\Omega)Y(e^{j\Omega T}) = \begin{cases} TY(e^{j\Omega T}), & |\Omega| < \dfrac{\pi}{T} \\ 0, & otherwise \end{cases}$$

CONTINUOUS-TIME PROCESSING OF DISCRETE-TIME SIGNALS

- C/D Converter
  - Output of C/D Converter

$$x[n] = x_c(nT)$$

$$X(e^{jw}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{w}{T} - \frac{2\pi k}{T}\right)\right)$$

DISCRETE-TIME PROCESSING OF CONTINUOUS-TIME SIGNALS

- D/C Converter
  - Output of D/C Converter

$$y_r(t) = \sum_{n=-\infty}^{\infty} y[n] \frac{\sin\left(\pi\left(t - nT\right)/T\right)}{\pi\left(t - nT\right)/T}$$

$$Y_r(j\Omega) = H_r(j\Omega) Y\left(e^{j\Omega T}\right)$$

$$= \begin{cases} TY\left(e^{j\Omega T}\right), & |\Omega| < \dfrac{\pi}{T} \\ 0, & otherwise \end{cases}$$

$$H_r(j\Omega) = \begin{cases} T, & |\Omega| < \dfrac{\pi}{T} \\ 0, & otherwise \end{cases}$$

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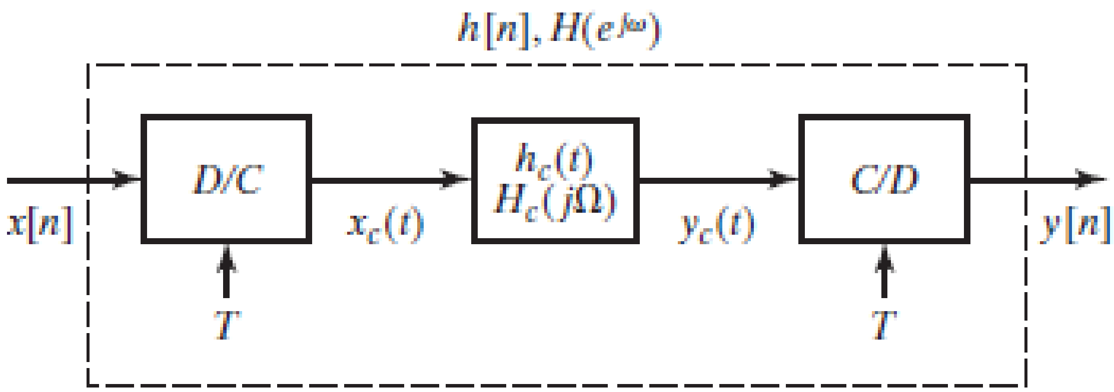
CONTINUOUS-TIME PROCESSING OF DISCRETE-TIME SIGNALS



$$x_c(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin[\pi(t - nT)/T]}{\pi(t - nT)/T}$$

$$y_c(t) = \sum_{n=-\infty}^{\infty} y[n] \frac{\sin[\pi(t - nT)/T]}{\pi(t - nT)/T},$$

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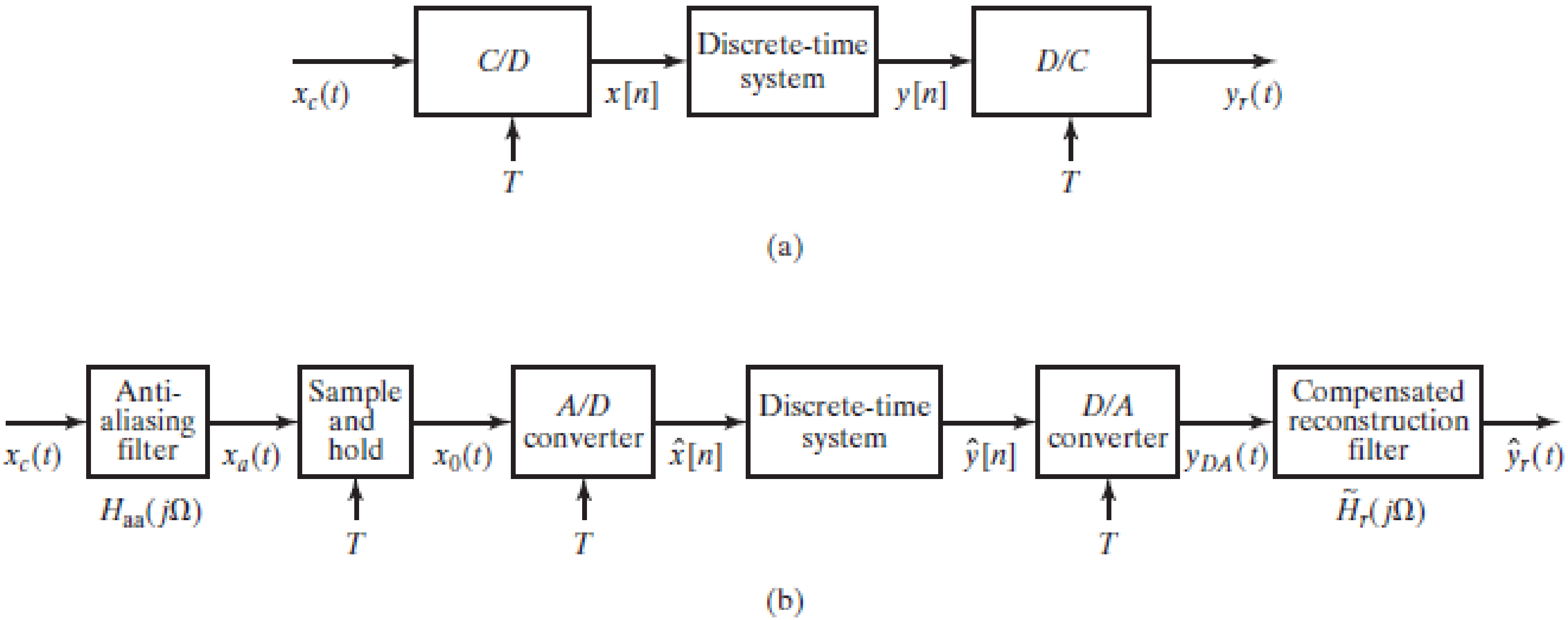


Figure 4.47 (a) Discrete-time filtering of continuous-time signals. (b) Digital processing of analog signals.

DIGITAL PROCESSING OF ANALOG SIGNALS

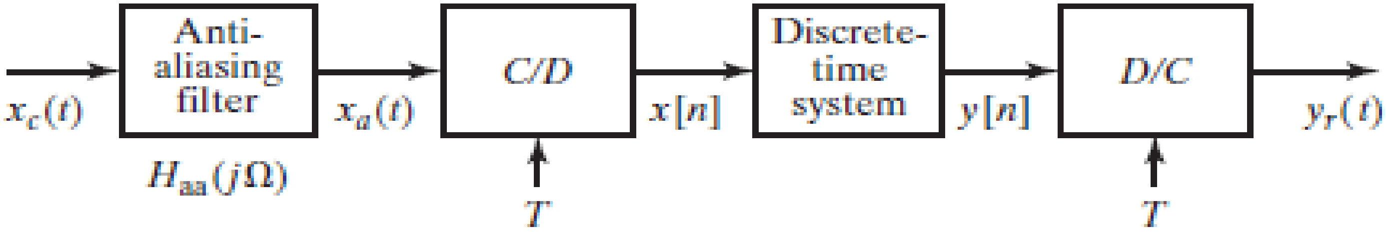


Figure 4.48 Use of prefiltering to avoid aliasing.

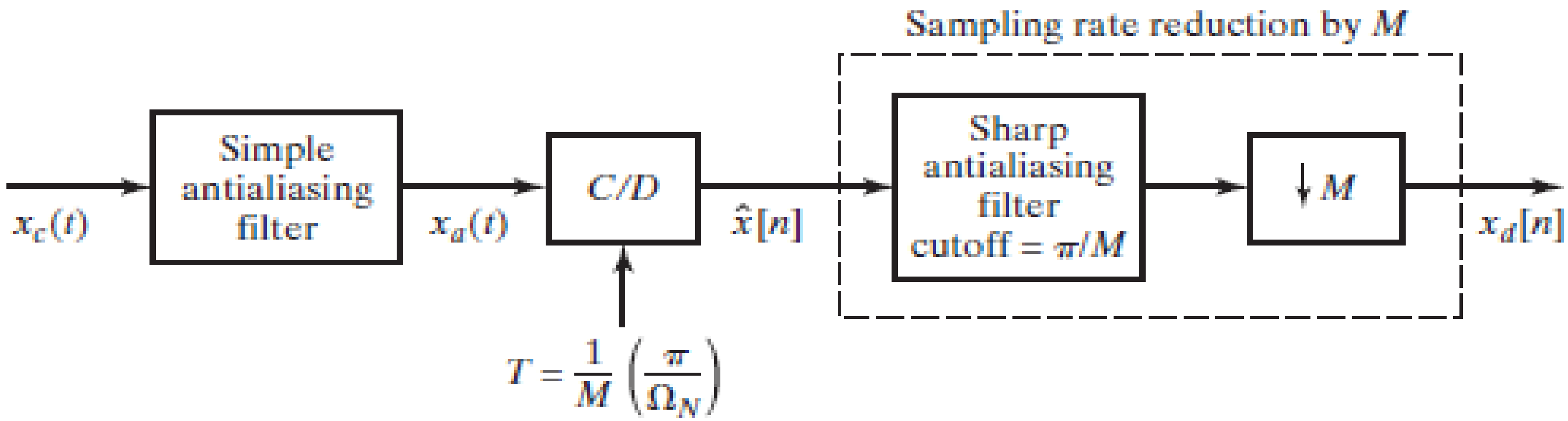
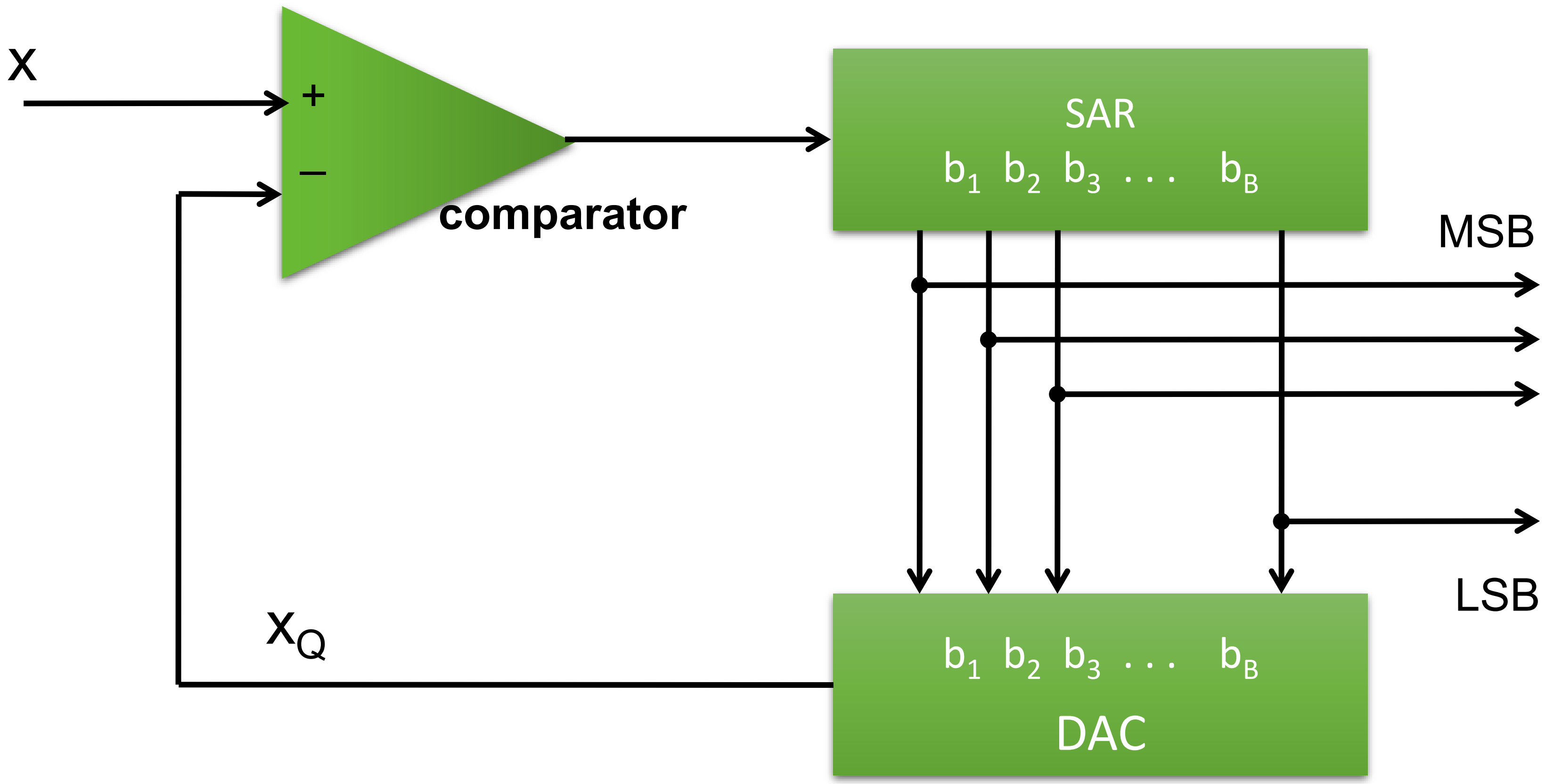


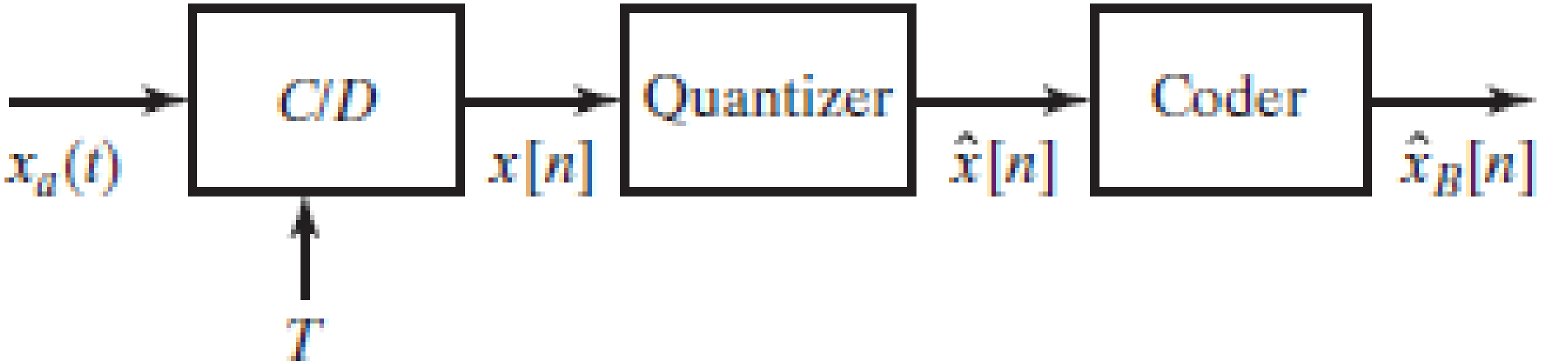
Figure 4.49 Using oversampled A/D conversion to simplify a continuous-time antialiasing filter.

DIGITAL PROCESSING OF ANALOG SIGNALS

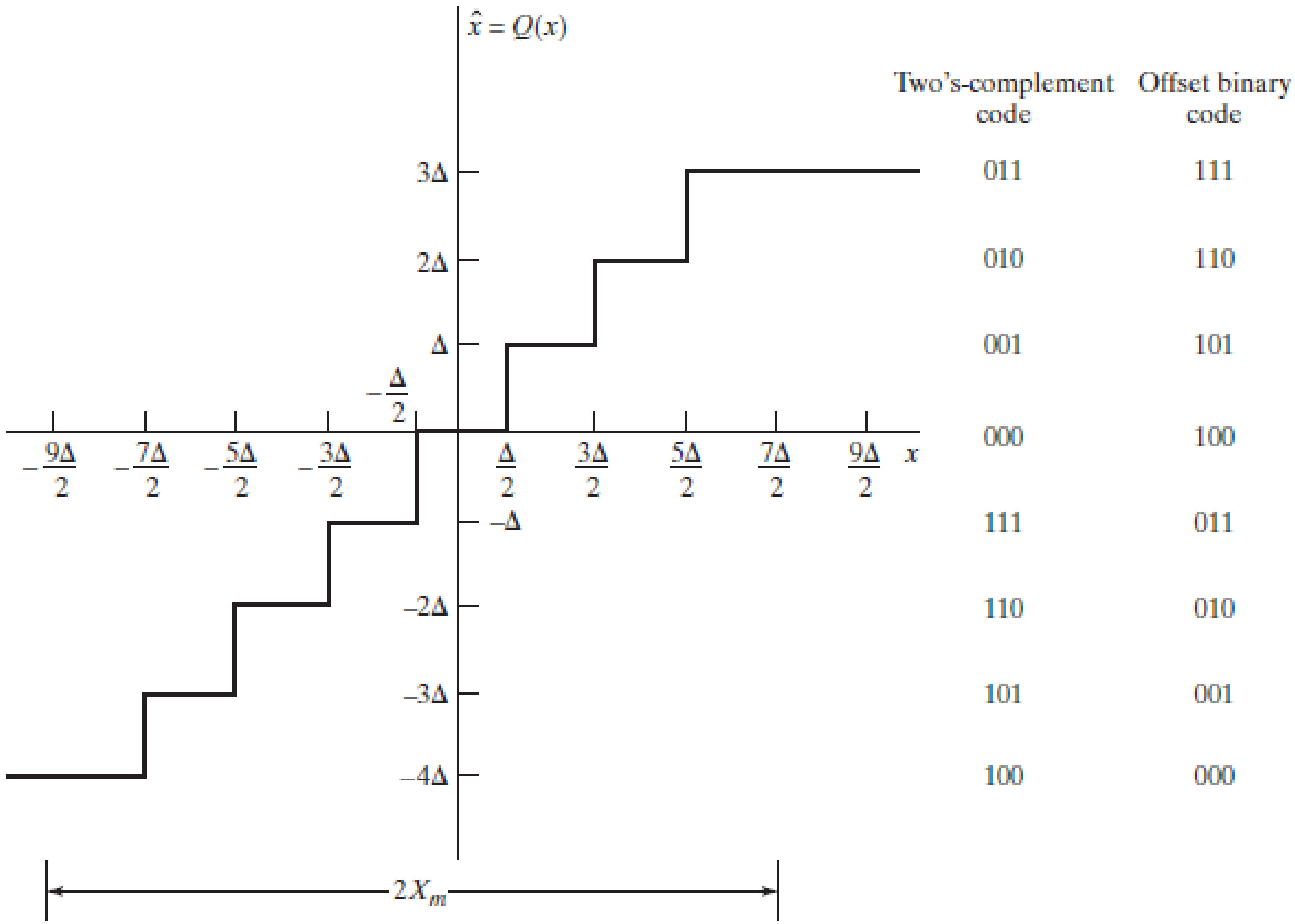


DIGITAL PROCESSING OF ANALOG SIGNALS

$$\hat{x}[n] = Q(x[n])$$



DIGITAL PROCESSING OF ANALOG SIGNALS



- Unipolar natural binary

$$x_Q = X_m(b_1 2^{-1} + b_2 2^{-2} + \ldots + b_B 2^{-B})$$

- Bipolar offset binary

$$x_Q = X_m(b_1 2^{-1} + b_2 2^{-2} + \ldots + b_B 2^{-B} - 0.5)$$

- Bipolar 2's complement

$$x_Q = X_m(\bar{b}_1 2^{-1} + b_2 2^{-2} + \ldots + b_B 2^{-B} - 0.5)$$

- Example: Quantized x = 3.5 by performing Unipolar natural binary

Test	$b_1b_2b_3b_4$	x_Q	$C = u(x - x_Q)$
b_1	1000	5,000	0
b_2	0100	2,500	1
b_3	0110	3,750	0
b_4	0101	3,125	1
	0101	3,125	

=> b = [0101]

- **Quantization Errors**

$$e_{rms} = \sqrt{\overline{e}^2} = \frac{\Delta}{\sqrt{12}}$$

$$e_{\max} = \Delta / 2$$

Signal to noise ratio $SNR = \frac{X_m}{\Delta}$

Change to dB : $SNR = 6B(dB)$

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THANK YOU